Available online at www.sciencedirect.com
ScienceDirect
Journal of Transport Geography 16 (2008) 126–133
JOURNAL OF TRANSPORT GEOGRAPHY
www.elsevier.com/locate/jtrangeo

Combining RP and SP data: biases in using the nested logit 'trick'– contrasts with flexible mixed logit incorporating panel and scale effects

David A. Hensher [a,*], John M. Rose [a], William H. Greene [b]

[a] Faculty of Economics and Business, Institute of Transport and Logistics Studies, The University of Sydney, NSW, Sydney 2006, Australia
[b] Stern School of Business, New York University, United States

Abstract

It has become popular practice that joint estimation of choice models that use stated preference (SP) and revealed preference (RP) data requires a way of adjusting for scale to ensure that parameter estimates across data sets are not confounded by differences in scale. The nested logit 'trick' presented by Hensher and Bradley in 1993 [Hensher, D.A., Bradley, M., 1993. Using stated response data to enrich revealed preference discrete choice models. Marketing Letters 4 (2), 139–152] continues to be widely used, especially by practitioners, to accommodate scale differences. This modelling strategy has always assumed that the observations are independent, a condition of all GEV models, which is not strictly valid within a stated preference experiment with repeated choice sets and between each SP observation and the single RP data point. This paper promotes the replacement of the NL 'trick' method with an error components model that can accommodate correlated observations as well as reveal the relevant scale parameter for subsets of alternatives. Such a model can also incorporate "state" or reference dependence between data types and preference heterogeneity on observed attributes. An example illustrates the difference in empirical evidence.

Keywords: Combined SP and RP data; Nested logit trick; Scale parameters; Error component mixed logit; Reference dependence

1. Introduction

It is common practice in estimation of many choice models which combine multiple data sources (e.g., RP and SP data sets) to use a nested logit (NL) structure as a 'trick or mechanism to reveal differences in scale between data sources' (Bradley and Daly, 1992, 1997; Hensher and Bradley, 1993). It is a trick in the sense that the underlying conditions to comply with utility maximisation such as the 0–1 bound on the inclusive value variable linking two levels in a nest (McFadden, 1981), while applicable between alternatives within SP and within RP choice sets, are not relevant *between* data sets – the scale differences between data sets (typically normalising to unity on one data source) is the only agenda.

In the majority of NL applications, the predictability of the set of NL structures studied is driven by the revelation of differences in SP–RP scale parameters and/or the partitioning of alternatives within a given data set in what is best described as 'commonsense' or intuitive partitions; for example, the marginal choice between car and public transport and then the choice between bus and train, conditional on choosing public transport. Hensher (1999) generalised the role of scale parameters through the use of the HEV search engine to allow for differences in scale, not only between data sets but between alternatives within and between data sets.

The NL model is a member of GEV models (McFadden, 1981) which cannot accommodate a number of specification

* Corresponding author. Tel.: +61 2 93510071; fax: +61 2 93510088.
E-mail addresses: Davidh@itls.usyd.edu.au (D.A. Hensher), Johnr@itls.usyd.edu.au (J.M. Rose), wgreene@stern.nyu.edu (W.H. Greene).

0966-6923/$ - see front matter © 2007 Elsevier Ltd. All rights reserved.
doi:10.1016/j.jtrangeo.2007.07.001

requirements of data that has repeated observations from the same respondent. This occurs with SP choice sets which exhibit potential correlation due to repeated observations or panel data. We need mixing of some kind for this, using the GEV model as a kernel (see Hess et al., 2005; Hess and Rose, 2006; Garrow and Bodea, 2005).

In addition to potential observation correlation, joint RP–SP estimation induces a potential 'state' or reference dependence effect, defined as the influence of the actual (revealed) choice on the stated choices of the individual. Reference dependence can manifest itself as a positive or negative effect of the choice of an alternative on the utility associated with that alternative in the stated responses (Bhat and Castelar, 2002). In a real sense it is a reflection of accumulated experience and the role that reference dependency plays in choosing, in the spirit of prospect theory (Kahneman and Tversky, 1979; Hensher, 2006; Hess et al., in press).

It is possible that the effect of reference dependence is positive for some individuals and negative for others (see Ailawadi et al., 1999), suggesting that an unconstrained analytical distribution for the random parameterisation of state dependence is appropriate. A positive effect may be the result of habit persistence, inertia to explore another alternative, or learning combined with risk aversion. A negative effect could be the result of variety seeking or the result of latent frustration with the inconvenience associated with the currently used alternative (Bhat and Castelar, 2002).

Thus, joint RP–SP estimation should not only recognise state dependence, but also accommodate heterogeneity in the reference dependence effect if it exists. Most RP–SP studies disregard reference dependence and adopt fixed parameters (i.e., homogeneity of attribute preference). Bhat and Castelar (2002) accommodate such unobserved heterogeneity in the reference dependence effect of the RP choice on SP choices. Brownstone et al. (1996), on the other hand, accommodate observed heterogeneity in the reference dependence effect by interacting the RP choice dummy variable with socio-demographic characteristics of the individual and SP choice attributes.

The paper outlines a very general mixed logit model which brings together the many recent contributions in the literature that deliver a flexible structure that can account for between-alternative error structure including correlated choice sets, RP–SP scale difference, unobserved preference heterogeneity, and reference dependency. An empirical example is used to illustrate the behavioural differences between the traditional NL-trick model and the flexible mixed logit model. We recognise parallel developments in GEV mixture models as summarised and extended by Hess et al. (2005), that can capture scale differences through an underlying NL model while also capturing other phenomena addressed herein through random terms.

2. The mixed logit framework

We begin with the basic form of the multinomial logit model, with alternative-specific constants α_{ji} and attributes x_{ji}, for individuals $i = 1, \ldots, N$ in choice setting t

$$\text{Prob}(y_{it} = j_t) = \frac{\exp(\alpha_{ji} + \boldsymbol{\beta}_i' \mathbf{x}_{jit})}{\sum_{q=1}^{J_i} \exp(\alpha_{qi} + \boldsymbol{\beta}_i' \mathbf{x}_{qit})}. \tag{1}$$

The random parameter model emerges as the form of the individual specific parameter vector, $\boldsymbol{\beta}_i$ is developed. The most familiar, simplest version of the model specifies

$$\begin{aligned} \beta_{ki} &= \beta_k + \sigma_k v_{ik} \text{ and} \\ \alpha_{ji} &= \alpha_j + \sigma_j v_{ji}, \end{aligned} \tag{2}$$

where β_k is the population mean for the kth attribute $(k = 1, \ldots, K)$, v_{ik} is the individual specific heterogeneity, with mean zero and standard deviation one, and σ_k is the standard deviation of the distribution of β_{ik}'s around β_k. The term 'mixed logit' is increasingly used in the literature (e.g., Revelt and Train, 1998; Train, 2003; Hensher et al., 2005) for this model. The choice specific constants, α_{ji} and the elements of $\boldsymbol{\beta}_i$ are distributed randomly across individuals with fixed means.

The $v_{ij'}$s are individual and choice specific, unobserved random disturbances – the source of the heterogeneity. For the full vector of K random coefficients in the model, we may write the full set of random parameters as

$$\boldsymbol{\rho}_i = \boldsymbol{\rho} + \boldsymbol{\Gamma} \mathbf{v}_i, \tag{3}$$

where $\boldsymbol{\Gamma}$ is a diagonal matrix which contains σ_k on its diagonal. We can allow the random parameters to be correlated. An additional layer of individual heterogeneity may be added to the model in the form of the error components that capture influences that are related to alternatives in contrast to attributes. We do this by constructing a set of independent individual terms, E_{im}, $m = 1, \ldots, M \sim N[0,1]$ that can be added to the utility functions. This device allows us to create what amounts to a random effects model and, in addition, a very general type of nesting of alternatives. Let θ_m be the scale parameter (standard deviation) associated with these effects. Then, each utility function can be constructed as

$$U_{ijt} = \alpha_{ji} + \beta_i' x_{jit} + (\text{any of } \theta_1 E_{i1}, \theta_2 E_{i2}, \ldots, \theta_M E_{iM}). \tag{4}$$

Consider, for example, a four outcome structure

$$\begin{aligned} U_{i1t} &= V_{i1t} + \theta_1 E_{i1} + \theta_2 E_{i2}, \\ U_{i2t} &= V_{i2t} + \theta_2 E_{i2}, \\ U_{i3t} &= V_{i3t} + \theta_1 E_{i1} + \theta_3 E_{i3}, \\ U_{i4t} &= V_{i4t} + \theta_4 E_{i4}. \end{aligned}$$

Thus, U_{i4t} has its own uncorrelated effect, but there is a correlation between U_{i1t} and U_{i2t} and between U_{i1t} and U_{i3t}. This example is fully populated, so the covariance matrix is block diagonal with the first three freely correlated. The model might usefully be restricted in a specific application. A convenient way to allow different structures is to introduce the binary variables $d_{jm} = 1$ if random term E_m appears in utility function j and zero otherwise. Then, the entire model can be specified as

$$\text{Prob}(y_{it} = j) = \frac{\exp\left[\alpha_{ji} + \beta_i' x_{jit} + \Sigma_{m=1}^{M} d_{jm}\theta_m E_{im}\right]}{\sum_{q=1}^{J_i} \exp[\alpha_{qi} + \beta_i' x_{qit} + \Sigma_{m=1}^{M} d_{qm}\theta_m E_{im}]}. \tag{5}$$

The model also contains individual alternative-specific constants α_{ji} and taste parameters β_i. This accommodates the individual heterogeneity in individual preferences with respect to the choices and the attributes. The randomness in these parameters is captured in a possibly heterogeneous distribution,

$$\boldsymbol{\beta}_i = \boldsymbol{\beta} + \mathbf{w}_i, \tag{6}$$

where w_i has mean vector 0 and covariance matrix Σ. The constants α_{ji} are parameterized likewise. It is convenient to introduce the variance parameters for the random parameters explicitly, which we do by letting v_i be a vector of independent random variables with mean zero and variance one, and

$$\boldsymbol{\beta}_i = \boldsymbol{\beta} + \boldsymbol{\Gamma\Omega}\mathbf{v}_\mathrm{i}, \tag{7}$$

where Ω is a diagonal matrix of standard deviations and Γ is a lower triangular matrix that allows correlation across parameters. If the random parameters are uncorrelated, then $\Gamma = \mathrm{I}$. The notation is simplified if we include the alternative-specific constants in β_i and the corresponding choice specific dummy variables in x_{jit}. The conditional choice probabilities are then obtained by inserting the expression for β_i in the utility functions, to obtain (based on Greene and Hensher, 2007)

$$\text{Prob}(y_{it} = j | v_i, E_i) = \frac{\exp\left[(\beta + \boldsymbol{\Gamma\Omega} v_i) x_{jit} + \Sigma_{m=1}^{M} d_{jm}\theta_m E_{im}\right]}{\sum_{q=1}^{J_i} \exp\left[(\beta + \boldsymbol{\Gamma\Omega} v_i) x_{qit} + \Sigma_{m=1}^{M} d_{qm}\theta_m E_{im}\right]}. \tag{8}$$

Because the conditional choice probabilities are functions of the unobserved individual specific random terms, they cannot be used to form the likelihood function for estimation of the parameters. The unconditional choice probabilities are formed by integrating the heterogeneity out of the conditional ones. Thus

$$\text{Prob}(y_{it} = j) = \int_{v_i} \int_{E_i} \text{Prob}(y_{it} = j | v_i, E_i) f(v_i, E_i)\, \mathrm{d}E_i\, \mathrm{d}v_i. \tag{9}$$

The integral does not exist in closed form, so we have approximated it with simulation (see Bhat (2003), Revelt and Train (1998), Train (2003), Brownstone et al. (2000) for details). The simulated, conditional choice probabilities are

$$\text{Simulated Prob}(y_{it} = j) = \frac{1}{R}\sum_{r=1}^{R} \frac{\exp\left[(\beta + \boldsymbol{\Gamma\Omega} v_{ir}) x_{jit} + \Sigma_{m=1}^{M} d_{jm}\theta_m E_{im,r}\right]}{\sum_{q=1}^{J_i} \exp\left[(\beta + \Gamma\Omega v_{ir}) x_{qit} + \Sigma_{m=1}^{M} d_{qm}\theta_m E_{im,r}\right]}, \tag{10}$$

where v_{ir} and E_{ir} are random draws on v_i and E_i, both of which are standard normally distributed. The simulated log-likelihood for n individuals and T_i choices by each individual is then

$$\ln L_i^S = \sum_{i=1}^{n} \ln\left\{\frac{1}{R}\sum_{r=1}^{R}\prod_{t=1}^{T_i} \frac{\exp\left[(\beta + \boldsymbol{\Gamma\Omega} v_{ir}) x_{jit} + \Sigma_{m=1}^{M} d_{jm}\theta_m E_{im,r}\right]}{\sum_{q=1}^{J_i} \exp\left[(\beta + \boldsymbol{\Gamma\Omega} v_{ir}) x_{qit} + \Sigma_{m=1}^{M} d_{qm}\theta_m E_{im,r}\right]}\right\}. \tag{11}$$

Some specific features of the model of interest in joint estimation with multiple data sets are the possibility of 'state (reference) dependence' engendered in the SP data as a derivative of an RP market context; and the differences in the scale parameters for the SP data relative to the RP data. Formally, reference dependence is defined as (Bhat and Castelar, 2002)

$$\varphi_q(1 - \kappa_{qt,\mathrm{RP}}), \tag{12}$$

where $\kappa_{qt,\mathrm{RP}} = 1$ if an RP obs, 0 otherwise, and φ_q is the parameter estimate of reference dependence which can be fixed or random. This variable enters the utility expression for each SP alternative, with the capability to select a generic specification.

The scale parameter for one data set (or set of alternatives) relative to the other set, the latter normalized to 1.0, is obtained through the introduction in one data set, the SP data[1], of a set of alternative-specific constants (ASCs) that have a zero mean and free variance (Brownstone et al., 2000). The scale parameter is calculated using the formula in

$$\lambda_{qt} = [(1 - \kappa_{qt,\mathrm{RP}})\lambda] + \kappa_{qt,\mathrm{RP}}, \tag{13}$$

where λ is inversely proportional to the estimated standard deviation of the ASC of an alternative, according to the EV1 distribution, where $\lambda = \pi/\sqrt{6}$Std Dev =1.28255/Std Dev of ASC.

This model with error components for each alternative is identified. We are estimating the $\boldsymbol{\theta}$ parameters as if they were weights on attributes, not scales on disturbances. The parameters are identified in the same way that the $\boldsymbol{\theta}$s on the attributes are identified. The parameter on the error component is $(\delta_m \sigma_m)$, where σ_m is the standard deviation. Since the scale is not identified, we normalize it to one for estimation purposes, with the understanding that the sign and magnitude of the weights on the error components are carried by θ. The sign of δ_m is also not identified, since the same set of model results will emerge if the sign of every draw on the component were reversed – the estimator of δ

[1] We select the SP data set in the empirical application but the RP data set could have been chosen.

would simply change sign with them. Hence we can normalize the sign to plus, and estimate $|\delta_m|$, with the sign and the value of σ_m also normalized for identification purposes.

3. The data

In this section we illustrate the ideas discussed above using commuter mode choice data. The data were drawn from a combined revealed preference (RP) and stated choice experiment that was conducted in six Australian capital cities: Sydney, Melbourne, Brisbane, Adelaide, Perth and Canberra (Hensher et al., 2005). The universal choice set comprised the currently available modes plus two 'new' modes, light rail and bus-based transitway (often referred to as a busway). Respondents evaluated scenarios describing ways to commute between their current residence and workplace locations using different combinations of policy-sensitive attributes and levels. The purpose of the exercise was to observe and model their observed coping strategies in each scenario.

To obtain some spatial representation, we first established the trip length in terms of travel time that is relevant for the respondent's current commuting trip so that the travel choices can be given in a context which has some reality for the respondent. The travel choice sets are divided into trip lengths of (i) less than 30 min, (ii) 30–45 min and (iii) over 45 min.

For the stated choice experiment, four alternatives appeared in each travel choice scenario: (a) car (drive alone), (b) car (ride share), (c) bus or busway, and (d) train or light rail. Twelve types of showcards described scenarios involving combinations of trip length (3) and public transport pairs (4): bus vs. light rail, bus vs. train (heavy rail), busway vs. light rail, and busway vs. train. Appearance of public transport pairs in each card shown to respondents was based on an experimental design.

Five three-level attributes were used to describe public transport alternatives: (a) total in-vehicle time, (b) frequency of service, (c) closest stop to home, (d) closest stop to destination, and (e) fare. The attributes of the car alternatives were: (a) travel time, (b) fuel cost, (c) parking cost, (d) travel time variability, and for toll roads (e) departure times and (f) toll charge. The design allows orthogonal estimation of alternative-specific main effect models for each mode option: (a) car no toll, (b) car toll road, (c) bus, (d) busway, (e) train, and (f) light rail. The attribute ranges are summarised in Table 1.

The master design for the travel choice task was a 27×3^{30} orthogonal fractional factorial, which produced 81 scenarios or choice sets. The 27-level factor was used to block the design into 27 versions each with three choice sets containing two alternatives. Versions were balanced such that each respondent saw every level of each attribute exactly once. The 3^{30} portion of the master design is an orthogonal main effects design, which permits independent estimation of all effects of interest. Two 2-level attributes were used to describe bus/busway and train/light rail modes, such that bus/train options appear in 36 scenarios and busway/light rail in 45 scenarios.

In addition to the stated preference data, each respondent provided details of a current commuting trip for the chosen mode and one alternative mode. This enabled us to estimate a joint SP and revealed preference (RP) model of choice of mode for the journey to work. The data and detailed descriptions of the sampling process and data profile are provided in Hensher et al. (2005).

A face to face interview was undertaken with a member of each sampled household who was responsible for making decisions on where the household lives and what vehicle(s) they purchase. The home interview took approximately 45 min to be completed. In addition, a commuter questionnaire was left for the respondent to complete and collected three days later. The number of sampled households in each capital city is: Sydney (300), Melbourne (300), Brisbane (250), Adelaide (200), Perth (200) and Canberra (150). The final modal shares for the pooled data for RP and stated choice data are summarised in Table 2. The 1400 household interviews were reduced to 1187 commuters after allowance for household with no workers and data editing, which gave 3599 SC observations.

4. Results

Table 3 reports the final models for the NL-trick model and the mixed logit (ML) models. The level of service variables are generic within the car and public transport (PT) modes in recognition of differences in marginal disutilities between car and PT attributes that are commonly reported in the wider literature, and travel cost is generic across all modes. Preference heterogeneity for each of these attributes is accounted for by random parameters. We investigated a large number of analytical distributions, including, normal, lognormal and triangular and found that the triangular distribution gave the best[2].

The reference dependence effect was treated as a random parameter and was also assessed as normal and triangular distributions. We were unable to establish any statistically significant influence of the actual (revealed) choice on the stated choices of the individual, reporting the constrained triangular results in Table 3.

The scale parameters for subsets of the SP alternatives in the ML model were found to be statistically significant and greater than one for the bus and train modes; although not

[2] The triangular distribution was first used for random coefficients by Revelt and Train (1998) and Train (2001), later incorporated into Train (2003). Hensher and Greene (2003) also used it and it is increasingly being used in empirical studies. Let c be the centre and s the spread. The density starts at $c - s$, rises linearly to c, and then drops linearly to $c + s$. It is zero below $c - s$ and above $c + s$. The mean and mode are c. The standard deviation is the spread divided by $\sqrt{6}$; hence the spread is the standard deviation times$\sqrt{6}$. The height of the tent at c is $1/s$ (such that each side of the tent has area $s \times (1/s) \times (1/2) = 1/2$, and both sides have area $1/2 + 1/2 = 1$, as required for a density). The slope is $1/s^2$. The mean weighted average elasticities were also statistically equivalent.

Table 1
The set of attributes and attribute levels in the commuter mode choice experiment

	Car no toll	Car toll road	Public transport	Bus	Train	Busway	Light rail
Short (<30 min)							
Travel time to work	15, 20, 25	10, 12, 15	Total time in the vehicle (one-way)	10, 15, 20	10, 15, 20	10,15,20	10,15,20
Pay toll if you leave at this time (otherwise free)	None	6–10, 6:30–8:30, 6:30–9	Frequency of service	Every 5, 15, 25	Every 5, 15, 25	Every 5, 15, 25	Every 5, 15, 25
Toll (one-way)	None	1, 1.5, 2	Time from home to your closest stop	Walk 5, 15, 25 Car/bus 4,6,8	Walk 5, 15, 25 Car/bus 4, 6, 8	Walk 5, 15, 25 Car/bus 4, 6, 8	Walk 5, 15, 25 Car/bus 4, 6, 8
Fuel cost (per day)	3, 4, 5	1, 2, 3	Time to your destination from the closest stop	Walk 5, 15, 25 Bus 4, 6, 8	Walk 5, 15, 25 Bus 4, 6, 8	Walk 5, 15, 25 Bus 4, 6, 8	Walk 5, 15, 25 Bus 4, 6, 8
Parking cost (per day)	Free, \$10, \$20	Free, \$10, \$20	Return fare	1, 3, 5	1, 3, 5	1, 3, 5	1, 3, 5
Time variability	0, ±4, ±6	0, ±1, ±2					
Medium (30–45 min)							
Travel time to work	30, 37, 45	20, 25, 30	Total time in the vehicle (one-way)	20, 25, 30	20, 25, 30	20, 25, 30	20, 25, 30
Pay toll if you leave at this time (otherwise free)	None	6–10, 6:30–8:30, 6:30–9	Frequency of service	Every 5, 15, 25	Every 5, 15, 25	Every 5, 15, 25	Every 5, 15, 25
Toll (one-way)	None	2, 3, 4	Time from home to your closest stop	Walk 5, 15, 25 Car/bus 4, 6, 8	Walk 5, 15, 25 Car/bus 4, 6, 8	Walk 5, 15, 25 Car/bus 4, 6, 8	Walk 5, 15, 25 Car/bus 4, 6, 8
Fuel cost (per day)	6, 8, 10	2, 4, 6	Time to your destination from the closest stop	Walk 5, 15, 25 Bus 4, 6, 8	Walk 5, 15, 25 Bus 4, 6, 8	Walk 5, 15, 25 Bus 4, 6, 8	Walk 5, 15, 25 Bus 4, 6, 8
Parking cost (per day)	Free, \$10, \$20	Free, \$10, \$20	Return fare	2, 4, 6	2, 4, 6	2, 4, 6	2, 4, 6
Time variability	0, ±7, ±11	0, ±2, ±4					
Long (>45 min)							
Travel time to work	45, 55, 70	30, 37, 45	Total time in the vehicle (one-way)	30, 35, 40	30, 35, 40	30, 35, 40	30, 35, 40
Pay toll if you leave at this time (otherwise free)	None	6–10, 6:30–8:30, 6:30–9	Frequency of service	Every 5, 15, 25	Every 5, 15, 25	Every 5, 15, 25	Every 5, 15, 25
Toll (one-way)	None	3, 4.5, 6	Time from home to your closest stop	Walk 5, 15, 25 Car/bus 4, 6, 8	Walk 5, 15, 25 Car/bus 4, 6, 8	Walk 5, 15, 25 Car/bus 4, 6, 8	Walk 5, 15, 25 Car/bus 4, 6, 8
Fuel cost (per day)	9, 12, 15	3, 6, 9	Time to your destination from the closest stop	Walk 5, 15, 25 Bus4, 6, 8	Walk 5, 15, 25 Bus 4, 6, 8	Walk 5, 15, 25 Bus 4, 6, 8	Walk 5, 15, 25 Bus 4, 6, 8
Parking cost (per day)	Free, \$10, \$20	Free, \$10, \$20	Return fare	3, 5, 7	3, 5, 7	3, 5, 7	3, 5, 7
Time variability	0, ±11, ±17	0, ±7, ±11					

All cost items are in \$'s, all time items are in minutes. In model estimation, fuel cost is converted to cents.

Table 2
Profile of modal share (%) for the RP and SC data

	RP	SC
Drive alone	59.2	28.5
Ride share	20.8	17.0
Bus	8.9	12.0
Train	11.1	11.9
Light rail	–	16.0
Busway	–	14.6
Total (number)	1187	3599

statistically significantly different from 1.0, the normalized value for the RP data. The car modes had a scale parameter of 0.035, suggesting a much higher variance on the unobserved effects associated with the EV1 random component; however it has a *t*-ratio of 0.70. What this suggests is that there is no serious violation of scale differences between the RP and SP data. This may be due in part to the capturing of relevant unobserved heterogeneity through attributes (i.e., random parameters) and alternatives (i.e., error components).

A number of alternative groupings of alternatives in the error components found that combining drive alone mode across the RP and SP data sets and the ride share gave statistically significant dispersion parameters than distinguishing RP and SP data. We found that separate error components for each of the car drive alone and car ride share alternatives across both RP and SP data sets gave statistically significant parameter estimate of 2.964 and 2.369 respectively in contrast to the fixed 1.0 for the full set of public modes (in RP and SP data sets). This suggests that there is substantial unobserved heterogeneity associated with the car alternatives, and especially the car drive alone alternative, that is greater than that associated with the public transport alternatives. This finding is intuitive given the large body of literature that suggests that the influencing attributes on car use are often more extensive (especially when including socio-demographic condition-

Table 3
Model results for 'nested logit' trick vs. panel mixed logit for combined revealed preference (RP) and stated preference (SP) choice data

Attribute	Alternative(s)	NL	Mixed logit (RP–EC panel)
In-vehicle cost	All	−0.5802 (−14.7)	R: −.9016 (−11.60)
Main mode time	RP and SP–DA, RS	−0.0368 (−6.4)	R: −0.1306 (−12.9)
Main mode time	RP and SP–BS, TN, LR, BWY	−0.0566 (−8.2)	R: −0.0745 (−8.03)
Access and egress mode time	RP and SP – BS, TN, LR, BWY	−0.0374 (−8.5)	R: −0.0570 (−9.74)
Reference Dependence	DA, RS, BS, TN	ns	R: −.0942 (−.79)
Personal income	RP and SP–DA	0.0068 (2.30)	0.0199 (3.98)
Drive alone constant	DA–RP	0.7429 (2.48)	2.5676 (7.65)
Ride share constant	RS–RP	−0.8444 (−3.1)	−0.6670 (−2.02)
Drive alone constant	DA–SP	0.0598 (0.36)	
Ride share constant	RS–SP	−0.2507 (−1.8)	
Train specific constant	TN–SP	0.1585 (1.40)	
Light rail specific constant	LR–SP	0.3055 (2.81)	
Busway specific constant	BWY–SP	−0.016 (−0.14)	
Bus specific constant	BS–RP	0.0214 (0.81)	0.0584 (0.21)
Random Parameter standard deviations:			
In-vehicle cost	All		1.803 (−11.6)
Main mode time	RP and SP–DA, RS		0.1384 (−7.27)
Main mode time	RP and SP–BS, TN, LR, BWY		0.0901 (−2.92)
Access and egress mode time	RP and SP–BS, TN, LR, BWY		0.0315 (−1.97)
Reference dependence	DA, RS, BS, TN		0.1326 (−.79)
SP to RP Scale Parameter	DA, RS		0.035 (0.70)
	BS, BWY		1.345 (7.11)
	TN, LR		1.271 (6.49)
Error Component (alternative-specific heterogeneity)	RP and SP–DA,		2.964 (12.7)
	RP and SP – RS		2.369 (10.1)
Scale parameters	RP and SP–DA, RS	1.00 (fixed)	
	RP and SP–BS, TN, LR, BWY	0.7321 (8.85)	
Sample size		2688	2688
Log-likelihood at convergence		−2668.1	−2313.27
Value of travel time savings:		$ Per person hour	
Main mode time	RP and SP–DA, RS	$3.81	$8.73 (1.43)
Main mode time	RP and SP–BS, TN, LR, BWY	$5.85	$4.96 (1.23)
Access and egress mode time	RP and SP–BS, TN, LR, BWY	$3.87	$3.79 (0.822)

The models were estimated in Nlogit 4[a] (cost is in dollars $AUD, time is in minutes) *t*-ratios in parenthesis.
Notes: R = random parameter mean estimates, DA = drive alone, RS – ride share, BS – bus, TN – train, LR – light rail, BWY – busway. We used 500 Halton draws to perform our integrations, so there is no simulation variance. Ns: the reference dependence term was extremely insignificant.
[a] Nlogit 4 was released for public access in May 2007 (see www.limdep.com).

ing) than the set that determine public transport use. Given the generally dominant role of the car in many cities (notably 70–85 percent modal share in Australian cities), one might expect greater preference heterogeneity in the car choosers and hence an increasing likelihood of greater unobserved heterogeneity. Interestingly the scale parameter in the NL model of 0.7321 for public transport suggests greater unobserved heterogeneity for public transport modes than the car; however while this may be the appropriate interpretation for this model, the absence of accounting for correlated choice sets, random preference heterogeneity in the attributes and in the alternatives makes the comparison somewhat trite.

Given the limitations of direct interpretation of parameter estimates from discrete choice models, behavioural contrasts are best made through outputs such as elasticities and willingness to pay estimates for specific attributes. The direct elasticities for in-vehicle cost and main mode time are given in Table 4, with values of travel time savings (VTTS) reported in Table 3. The VTTS for the mixed logit model are based on the conditional distributions (i.e., conditional on the alternative chosen)[3]. There are significant mean differences between the VTTS for main mode times for both car and public modes between the NL and ML models. There is significant under valuation for car under NL and slight over valuation for public modes. The access and egress VTTS for public modes are essentially the same.

The direct and cross elasticities for nested logit are relatively straightforward calculations in contrast to mixed logit (Train, 2003). The formulae differ in recognition of different substitution patterns amongst the alternatives. Mixed logit does not exhibit independence from irrelevant alternatives (IIA) or the relatively restrictive substitution patterns of nested logit. The ratio of mixed logit probabilities, Pni/Pnj, depends on all the data, including attributes of alternatives other than i or j. As shown by Train (2003), the denominators of the logit formula are inside the integrals and therefore do not cancel. The percentage change in the probability for one alternative given a change in the mth attribute of another alternative is

$$E_{ni_{x_{nj}^m}} = -\frac{1}{P_{ni}}\int \beta^m L_{ni}(\beta)L_{nj}(\beta)f(\beta)\,\mathrm{d}\beta,$$

where β^m is the mth element of β. This elasticity is different for each alternative i. A ten-percent reduction for one alternative need not imply (as with multinomial logit) a ten-percent reduction in each other alternative.

[3] A referee noted that the authors work with the distribution of the conditional means, which is different from the conditional distribution and which significantly underestimates the variance. In the past we have found that working with the full distribution is far from satisfactory. This is because it is still taking draws and averaging over these. In our experience this not only leads to increased variance but also leads to implausible willingness to pay values such as VTTS. Using the means of the distributions is statistically not ideal, doing so tends to yield much more plausible WTP results.

Table 4
Summary of direct choice elasticities for cost and in-vehicle time (mean and standard deviation across the sample)

Attribute	Alternative(s)	NL	Relativity	Mixed logit (RP–EC panel)
Invehicle cost				
	RP DA	*−.108 (.195)*	>	*−.081 (.114)*
	RP RS	*−.363 (.419)*	>	*−.251 (.293)*
	RP BS	*−.510 (.516)*	>	*−.483 (.469)*
	RPTN	*−.638 (.538)*	<	*−.653 (.532)*
	SP LR	*−.544 (.384)*	<	*−.588 (.409)*
	SP BWY	*−.575 (.389)*	>	*−.551 (.382)*
	SP DA	−.812 (.630)		−.556 (.406)
	SP RS	−.957 (.639)		−.700 (.509)
	SP BS	−.545 (.410)		−.534 (.409)
	SP TN	−.568 (.409)		−.608 (.432)
Main mode time				
	RP DA	*−.110 (.190)*	<	*−.209 (.223)*
	RP RS	*−.396 (.387)*	<	*−.611 (.509)*
	RP BS	*−.571 (.583)*	>	*−.443 (.430)*
	RPTN	*−.763 (.664)*	>	*−.655 (.542)*
	SP LR	*−.657 (.325)*	>	*−.601 (.266)*
	SP BWY	*−.618 (.354)*	>	*−.485 (.264)*
	SP DA	−.536 (.372)		−.753 (.393)
	SP RS	−.639 (.375)		−.935 (.458)
	SP BS	−.586 (.339)		−.475 (.262)
	SP TN	−.701 (.330)		−.616 (.269)

When we look at the direct elasticities, we get some significant differences. We focus only on the elasticities associated with the revealed preference alternatives where the alternatives currently exist, since the SP estimates are somewhat meaningless given that the probabilities in the elasticity formula are based on hypothetical choice responses and hence market shares[4]. The RP shares are calibrated market shares. For the new modes we have to rely on the SP estimates. On average there are some noticeable differences, especially for ride share where the mean cost elasticity is significantly lower for the UML in contrast to the NL-trick model; yet significantly higher for main mode time. Practitioners will use mean estimates only and indeed will obtain noticeably different market share predictions when using mean elasticity estimates from each model.

5. Conclusions

This paper has set out a more general discrete choice model than the still very popular nested logit specification used by many practitioners and researchers to account for scale differences between multiple data sets, commonly a revealed preference data set and a stated choice set. The nested logit approach is limiting in that is not capable of accounting for the potential correlation induced through

[4] SP derived elasticities make sense only when the alternative-specific constants associated with alternatives that currently exist in real markets are calibrated to reproduce base modal shares. This is required in order to obtain relevant choice probabilities, an input into the calculation of elasticities.

repeated observations on one or more pooled data sets. Nor does it recognize the role that various sources of heterogeneity play in influencing choices outcomes, either via the random parameterization of observed attributes and via parameterization of error components associated with a single or a sub-set of alternatives (alternative-specific heterogeneity).

The mixed logit model presented herein is capable of allowing for these influencing dimensions, (observed or unobserved) in addition to accounting for scale differences (that are equivalent to the scale revealed in the NL model). The empirical example illustrates the additional outputs from the ML and the differences in key behavioural outputs.

Acknowledgements

Discussions with Chandra Bhat, David Brownstone and Stephane Hess have been invaluable in the preparation of this paper as have comments from three referees.

References

Ailawadi, K.L., Gedenk, K., Neslin, S.A., 1999. Heterogeneity and purchase event feedback in choice models: an empirical analysis with implications for model buildings. International Journal of Research in Marketing 16, 177–198.

Bhat, C.R., 2003. Simulation estimation of mixed discrete choice models using randomized and scrambled Halton sequences. Transportation Research B 37 (9), 837–855.

Bhat, C.R., Castelar, S., 2002. A unified mixed logit framework for modeling revealed and stated preferences: formulation and application to congestion pricing analysis in the San Francisco Bay area. Transportation Research B 36, 577–669.

Bradley, M.A., Daly, A.J., 1992. Uses of the logit scaling approach in stated preference analysis. In: Paper Presented at the 7th World Conference on Transport Research, Lyon, July.

Bradley, M.A., Daly, A.J., 1997. Estimation of logit choice models using mixed stated preference and revealed preference information. In: Stopher, P.R., Lee-Gosselin, M. (Eds.), Understanding Travel Behaviour in an Era of Change. Pergamon, Oxford, pp. 209–232.

Brownstone, D., Bunch, D.S., Golob, T., Ren, W., 1996. A transactions choice model for forecasting demand for alternative-fuel vehicles. In: McMullen, S. (Ed.), Research in Transportation Economics, vol 4. JAI Press Inc., pp. 87–129.

Brownstone, D., Bunch, D.S., Train, K., 2000. Joint mixed logit models of stated and revealed preferences for alternative-fuel vehicles. Transportation Research 34B, 315–338.

Garrow, L.A., Bodea, T.D., 2005. A rigorous framework for empirically comparing mixed GEV and mixed MNL with error components models. In: Proceedings from the European Transport Conference, Strasbourg, France. pp. 1–20.

Greene, W.H., Hensher, D.A., 2007. Mixed Logit and Heteroscedastic Control for Random Coefficients and Error Components. Transportation Research E.

Hensher, D.A., 1999. HEV Choice Models as a Search Engine for Specification of Nested Logit Tree Structures. Marketing Letters 10 (4), 333–343.

Hensher, D.A., 2006. How do Respondents Handle Stated Choice Experiments? – Attribute processing strategies under varying information load. Journal of Applied Econometrics 21, 861–878.

Hensher, D.A., Bradley, M., 1993. Using stated response data to enrich revealed preference discrete choice models. Marketing Letters 4 (2), 139–152.

Hensher, D.A., Greene, W.H., 2003. Mixed logit models: state of practice. Transportation 30 (2), 133–176.

Hensher, D.A., Rose, J., Greene, W.H., 2005. Applied Choice Analysis: A Primer. Cambridge University Press, Cambridge.

Hess, S., Rose, J.M., 2006. Effects of Distributional Assumptions on Conditional Estimates from Mixed Logit Models. Institute of Transport and Logistics Studies, The University of Sydney.

Hess, S., Bierlaire, M., Polak, J., 2005. Capturing correlation and taste heterogeneity with mixed GEV models. In: Scarpa, Riccardo, Alberini, Anna (Eds.), Environmental and Resource Economics. Springer Publisher, Dordrecht, The Netherlands, pp. 55–76.

Hess, S., Rose, J.M., Hensher, D.A., Asymmetrical preference formation in willingness to pay estimates in discrete choice models. Transportation Research E, in press.

Kahneman, D., Tversky, A., 1979. Intuitive prediction: biases and corrective procedures. In: Makridakis, S., Wheelwright, S.C. (Eds.), Studies in the Management Sciences: Forecasting, vol. 12. North Holland, Amsterdam.

McFadden, D.L., 1981. In: Manski, C.F., McFadden, D.L. (Eds.), Econometric Models of Probabilistic Choice in Structural Analysis of Discrete Data. MIT Press, Cambridge Massachusetts, pp. 198–271.

Revelt, D., Train, K., 1998. Mixed Logit with repeated choices: households' choices of appliance efficiency level. Review of Economics and Statistics 80, 1–11.

Train, K., 2001. A comparison of hierarchical Bayes and maximum simulated likelihood for mixed logit. Working Paper, Department of Economics, University of California, Berkeley. <http://elsa.berkeley.edu/~train/compare.pdf>.

Train, K.E., 2003. Discrete Choice Methods with Simulation. Cambridge University Press, Cambridge.